MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 09** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

MATHS MANTHAN | $(5 \times 1 = 5$ Marks)

Q1. If the sun's elevation increases from $30°$ to $45°$, the length of the shadow of a tower: [1]

(A) increases (B) decreases (C) remains identical (D) doubles

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Justification: Shadow length $s = h\cot\theta$. Since $\cot 45° = 1 < \sqrt{3} = \cot 30°$, the shadow length $\boxed{\text{decreases}}$.

Q2. The angle of elevation of the top of a 15 m high tower from a point 15 m away from its foot on level ground is: [1]

(A) $30°$ (B) $45°$ (C) $60°$ (D) $90°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Justification: $\tan\theta = \dfrac{15}{15} = 1 \implies \theta = \boxed{45°}$.

Q3. A ladder rests against a vertical wall at an angle α to horizontal. Its foot is pulled away by distance p and slides down distance q making angle β. Then $\dfrac{p}{q}$ equals: [1]

(A) $\dfrac{\cos\beta - \cos\alpha}{\sin\alpha - \sin\beta}$ (B) $\dfrac{\cos\alpha - \cos\beta}{\sin\beta - \sin\alpha}$ (C) $\dfrac{\sin\alpha - \sin\beta}{\cos\beta - \cos\alpha}$ (D) 1

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Let ladder length be L. Initially ground distance $= L\cos\alpha$ and height $= L\sin\alpha$; finally ground distance $= L\cos\beta$ and height $= L\sin\beta$. Hence $p = L(\cos\beta - \cos\alpha)$ and $q = L(\sin\alpha - \sin\beta)$, so $\dfrac{p}{q} = \boxed{\dfrac{\cos\beta - \cos\alpha}{\sin\alpha - \sin\beta}}$.

Q4. From a point P on level ground, the angle of elevation of the top of a tower is $30°$. If the tower is 100 m high, the distance of point P from the foot of the tower is: [1]

(A) $100\sqrt{3}$ m (B) $\dfrac{100}{\sqrt{3}}$ m (C) $50\sqrt{3}$ m (D) 200 m

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $d = 100\cot 30° = \boxed{100\sqrt{3}\text{ m}}$.

Q5. **Assertion (A):** If a vertical pole 6 m high casts a shadow $6\sqrt{3}$ m long on the ground, then the sun's elevation is $60°$.

Reason (R): For a vertical pole of height h and shadow s, $\tan\theta = \dfrac{h}{s}$; here $\tan\theta = \dfrac{6}{6\sqrt{3}} = \dfrac{1}{\sqrt{3}} \implies \theta = 30°$. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D)

Justification: Assertion (A) claims $60°$, which is false because $\tan\theta = \dfrac{6}{6\sqrt{3}} = \dfrac{1}{\sqrt{3}} \implies \theta = 30°$. Reason (R) computes this correctly, so (A) is false and (R) is true.

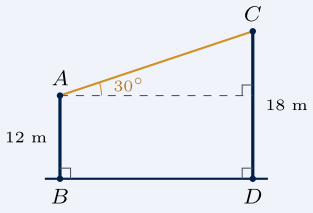
SECTION B: Very Short Answer Questions

MATHS MANTHAN | $(2 \times 2 = 4$ Marks)

Q6. The tops of two poles of heights 18 m and 12 m are connected by a wire. If the wire makes an angle of $30°$ with the horizontal, find the length of the wire. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let poles $AB = 12$ m and $CD = 18$ m be joined by wire AC, with vertical height difference $18 - 12 = 6$ m at $\angle = 30°$. [0.5 Mark]



$$\sin 30° = \frac{6}{AC} \implies \frac{1}{2} = \frac{6}{AC} \implies AC = \boxed{12\text{ m}}$$

Length of wire: [1.5 Marks]

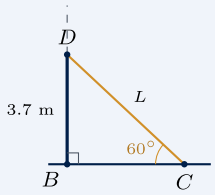
Q7. An electrician has to repair an electric fault on a pole of height 5 m. She needs to reach a point 1.3 m below the top using a ladder inclined at 60° to the horizontal. What should be the length of the ladder? [Take $\sqrt{3} = 1.732$] **[2]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

Reach height $BD = 5 - 1.3 = 3.7$ m, ladder CD inclined at 60° to ground BC. [0.5 Mark]

In right $\triangle DBC$: $\sin 60° = \dfrac{3.7}{CD} \implies CD = \dfrac{7.4}{\sqrt{3}} = \dfrac{7.4\sqrt{3}}{3}$ m. [1 Mark]

With $\sqrt{3} = 1.732$: $CD = \dfrac{7.4}{1.732} \approx \boxed{4.27 \text{ m}}$ $\left(\dfrac{7.4\sqrt{3}}{3} \text{ m}\right)$. [0.5 Mark]



SECTION C: Short Answer Questions — MATHS MANTHAN | (2 × 3 = 6 Marks)

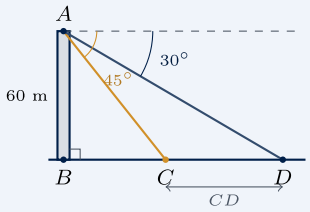
Q8. From the top of a chimney 60 m high, the angles of depression of two points on the ground on the same side are observed to be 30° and 45°. Find the distance between the two points. [Take $\sqrt{3} = 1.732$] **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let chimney $AB = 60$ m and ground points C, D on same side have depressions 45° and 30° ($\angle ACB = 45°, \angle ADB = 30°$). [0.5 Mark]

In $\triangle ABC$: $BC = 60 \cot 45° = 60$ m. [1 Mark]

In $\triangle ABD$: $BD = 60 \cot 30° = 60\sqrt{3}$ m. [1 Mark]



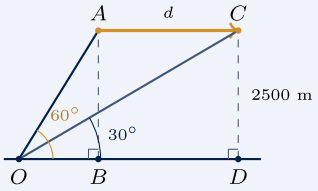
Distance between the two points $= CD = 60\sqrt{3} - 60 = 60(1.732 - 1) = 60 \times 0.732 = \boxed{43.92 \text{ m}}$. [0.5 Mark]

Q9. An aeroplane flying at a constant altitude of 2500 m is observed from a tracking station at an angle of elevation of 60°. After 20 seconds, the angle of elevation drops to 30°. Find the speed of the aeroplane in km/h. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let plane fly horizontally $A \to C$ at height 2500 m, observed from O at 60° and 30°. [0.5 Mark]

In $\triangle ABO$: $OB = \dfrac{2500}{\sqrt{3}}$ m; in $\triangle CDO$: $OD = 2500\sqrt{3}$ m. [1 Mark]



Distance flown $= BD = 2500\sqrt{3} - \dfrac{2500}{\sqrt{3}} = \dfrac{5000}{\sqrt{3}}$ m in 20 s. [1 Mark]

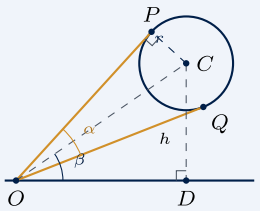
$$\text{Speed} = \frac{5000}{20\sqrt{3}} = \frac{250}{\sqrt{3}} \text{ m/s} = \frac{250}{\sqrt{3}} \times \frac{18}{5} = \frac{900}{\sqrt{3}} = 300\sqrt{3} = \boxed{300\sqrt{3} \text{ km/h} \approx 519.6 \text{ km/h}}$$

Evaluating speed in km/h: [0.5 Mark]

Q10. A round spherical balloon of radius r subtends an angle α at the eye of an observer while the angle of elevation of its centre is β. Prove that the height of the centre of the balloon above the horizontal observer plane is: $h = r \sin\beta \csc\dfrac{\alpha}{2}$. [5]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let O be the observer's eye and C be the centre of spherical balloon of radius r at height $CD = h$ above horizontal plane OD. Tangents OP, OQ from O touch sphere at P, Q, so $\angle POQ = \alpha$ and $\angle COD = \beta$. [1 Mark]



In $\triangle OPC$ and $\triangle OQC$: $CP = CQ = r$, $OC = OC$, and $\angle OPC = \angle OQC = 90°$ (radius $\perp$ tangent). Thus $\triangle OPC \cong \triangle OQC$ (RHS), giving $\angle POC = \angle QOC = \dfrac{\alpha}{2}$. [1.5 Marks]

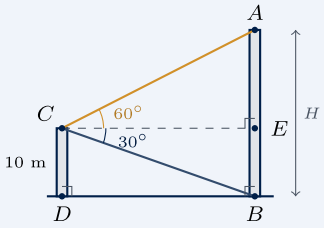
In right $\triangle OPC$: $\sin\dfrac{\alpha}{2} = \dfrac{CP}{OC} = \dfrac{r}{OC} \implies OC = r \csc\dfrac{\alpha}{2}$. [1.5 Marks]

In right $\triangle ODC$: $\sin\beta = \dfrac{CD}{OC} = \dfrac{h}{OC} \implies h = OC \sin\beta = \boxed{r \sin\beta \csc\dfrac{\alpha}{2}}$ [Hence Proved]. [1 Mark]

Q11. An observer standing on the deck of a ship, 10 m above water level, observes that the angle of elevation of the top of a hill is 60° and the angle of depression of the base of the hill is 30°. Calculate the distance of the hill from the ship and the total height of the hill. [5]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let ship deck be C at height $CD = 10$ m above water, and hill be AB. Horizontal line $CE \perp AB$ gives $EB = CD = 10$ m, $\angle ACE = 60°$, $\angle ECB = \angle CBD = 30°$. [1 Mark]



In right $\triangle CDB$: $\tan 30° = \dfrac{10}{DB} \implies DB = CE = 10\sqrt{3} = 10(1.732) = 17.32$ m. [1.5 Marks]

In right $\triangle AEC$: $\tan 60° = \dfrac{AE}{10\sqrt{3}} \implies AE = 10\sqrt{3} \times \sqrt{3} = 30$ m. [1.5 Marks]

Total height of hill $= AB = AE + EB = 30 + 10 = 40$ m.

$\therefore$ Distance of hill $= \boxed{10\sqrt{3} \text{ m} \approx 17.32 \text{ m}}$, Height of hill $= \boxed{40 \text{ m}}$

Final distance and height: [1 Mark]

Q12. CASE STUDY: Coastal Lighthouse Maritime Harbor Safety

A lighthouse beacon lamp L sits at effective height $LP = 120$ m above sea level. The keeper observes vessels A and B approaching along the same radial line from base P with angles of depression $60°$ (vessel A, closer) and $45°$ (vessel B, farther).

(i) Find the horizontal distance of vessel B from the base of the lighthouse (P). [1]

(ii) Find the horizontal distance of vessel A from the base of the lighthouse (P). [2]

(iii) Calculate the distance between vessel A and vessel B. [Take $\sqrt{3} = 1.732$] [2]

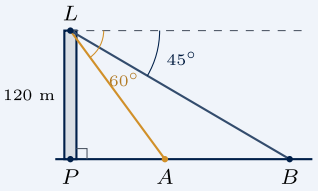
OR

If vessel A requires 2 minutes to cover the distance between vessel B and vessel A, find its average speed in m/s. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

(i) In right $\triangle LPB$ ($\angle LBP = 45°$):

$$\tan 45° = \frac{120}{PB} \implies PB = \boxed{120 \text{ m}}$$



Subpart (i): [1 Mark]

(ii) In right $\triangle LPA$ ($\angle LAP = 60°$):

$$\tan 60° = \frac{120}{PA} \implies PA = \frac{120}{\sqrt{3}} = \boxed{40\sqrt{3} \text{ m} \approx 69.28 \text{ m}}$$

Subpart (ii): [2 Marks]

(iii) Main Choice: Distance between A and B: $AB = PB - PA = 120 - 40\sqrt{3} = 120 - 69.28 = \boxed{50.72 \text{ m}}$. [2 Marks]

(iii) OR Choice: Time $t = 2$ minutes $= 120$ s. Average speed $= \frac{50.72}{120} = \boxed{0.4227 \text{ m/s} \approx 0.42 \text{ m/s}}$. [2 Marks]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★